SETUP ANALYSIS RESULTS

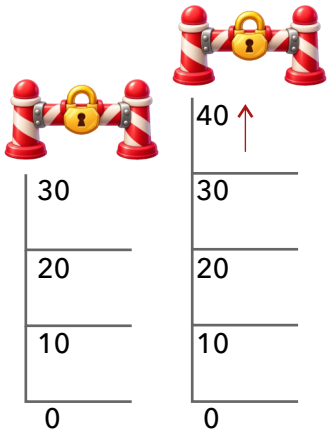
DOES GATE PLACEMENT AFFECT PLAYER RETENTION?



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QUESTION:



Will moving the first gate from level 30 to level 40 **improve 7-day player retention**?

#Data Source

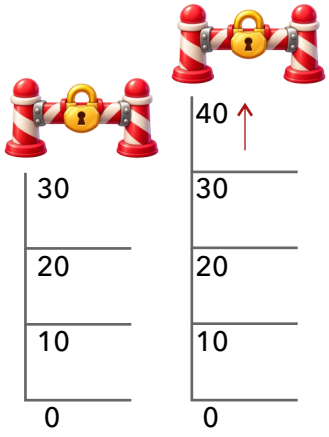
Dataset: *Mobile Games: A/B Testing*
Source: Kaggle



THE BORING STUFF

Cookie Cats uses progression gates that require players to wait or pay to continue.

- **Control:** the first gate is placed at level 30
- **Treatment:** the first gate is moved to level 40



BEFORE THE ANALYSIS:

- Check data quality
- Check for Sample Ratio Mismatch
- Compare player behavior and retention
- Estimate the effect and its uncertainty

Experiment	Question	Validation
Gate 30 → Gate 40	Does it improve 7-day retention?	Data quality + SRM

A/A test results and experiment dates are not available in the dataset.

DOES MOVING THE GATE AFFECT DAY 7 RETENTION?

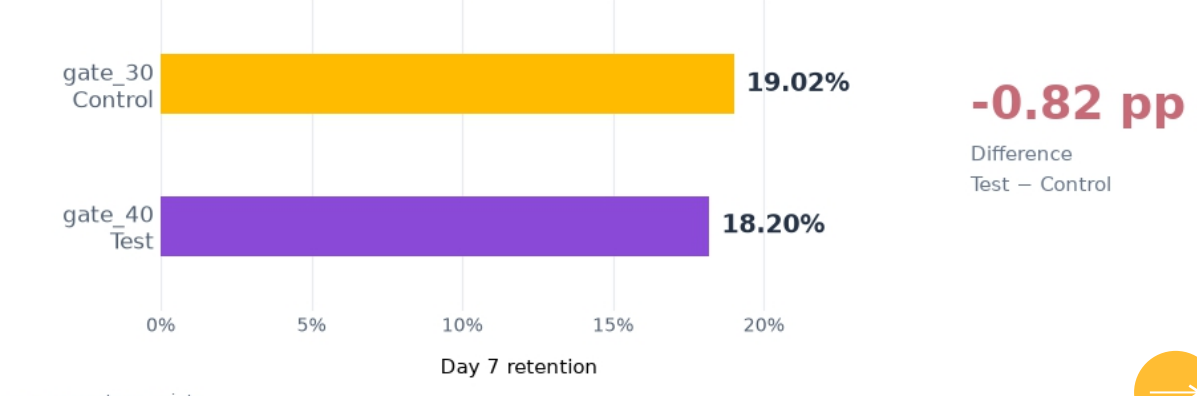
*Since historical data was unavailable, I used the control group's Day 7 retention as the baseline to estimate the MDE.

Group	Total users	Returned users	Retention
gate_30 — Control	44,700	8,502	19,02%
gate_40 — Test	45,489	8,279	18,20%

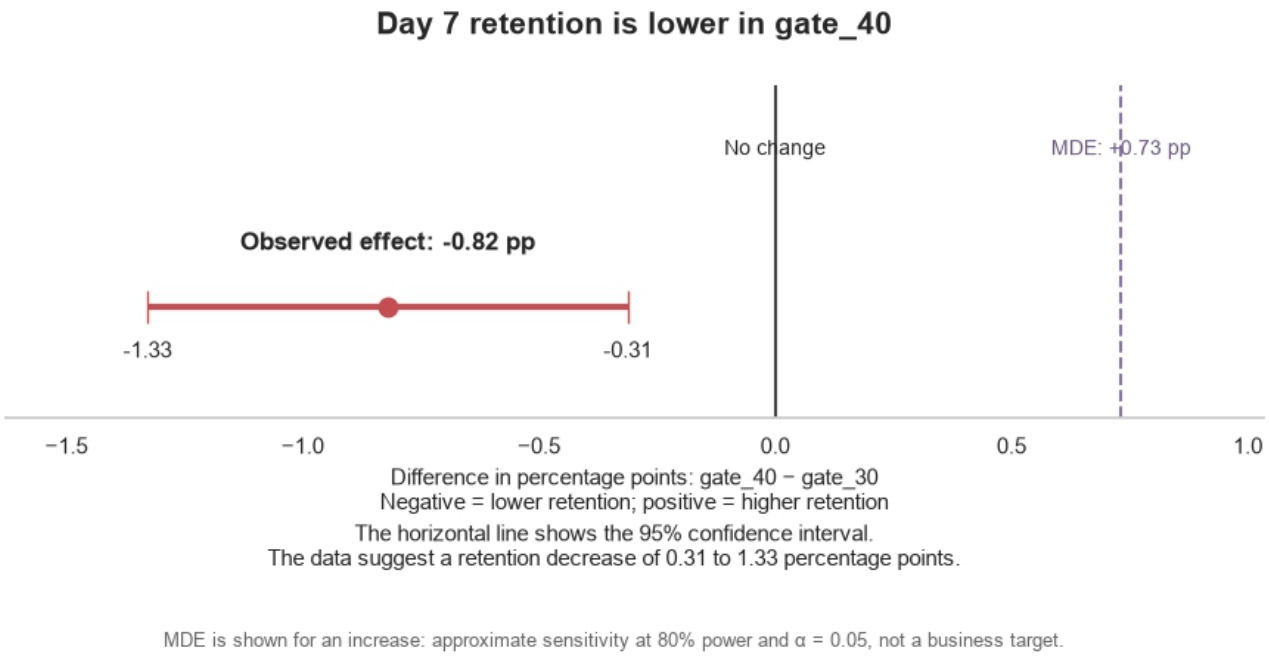
Difference	95% CI	p-value
−0.82 pp	[−1.33, −0.31] pp	0.0015

Approx. MDE	Relative MDE
0.73pp	3.85%

The p-value is below our significance level of 0.05, so I reject the null hypothesis.
I found a statistically significant difference between the control and test groups.



pp = percentage points



Day 7 retention decreased by 0.82 percentage points in the test group.
The entire 95% confidence interval is below zero, indicating a decrease in retention.

Since our goal was to improve Day 7 retention, this is a **negative result (■'red' test).**
❌ I do not recommend rolling out this feature.

Now, let's take a quick look at the other metrics



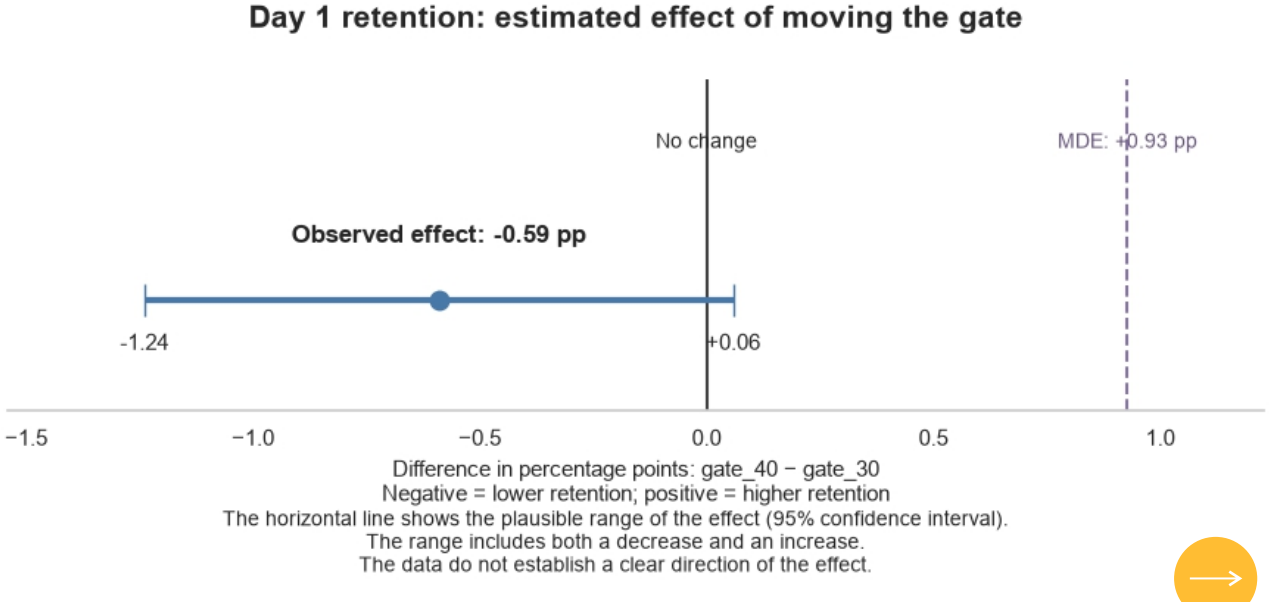
WHAT ABOUT DAY 1 RETENTION?

Group	Total users	Returned users	Retention
gate_30 — Control	44,700	20,034	44,82%
gate_40 — Test	45,489	20,119	44,23%

Difference	95% CI	p-value
−0.59 pp	[−1,24; +0,06] pp	0,0744

Approx. MDE	Relative MDE
0.93pp	2.07%

p-value > α, so we fail to reject H_0. Observed retention decreased, but the **95% CI includes zero: the result is inconclusive.**



MDE is shown for an increase: approximate sensitivity at 80% power and α = 0.05, not a business target.

AND FINALLY: MEAN GAME ROUNDS PER USER

T-statistic	Mean difference	p-value
– 0.0634	– 0.0433 rounds	0.9495

p-value > α, so we found no statistically significant difference between the control and test groups.

RECOMMENDATION:

Don’t roll out this feature in its current form:
it significantly reduced Day 7 retention, our primary metric.

